

G2F 2023 Weather Data

Weather data was collected at 30-minute intervals with a Spectrum WatchDog 2700 weather station throughout the growing season. Each station measured wind speed, direction, and gust; air temperature, dewpoint, and relative humidity; soil temperature and moisture; rainfall and solar radiation. Some stations had additional measurements including soil electrical conductivity, ultra-violet (UV) light, and photosynthetically active radiation (PAR).

The weather stations were equipped with several ports to attach external sensors. The data from the sensors were then collected in columns with headers that corresponded to the ports used. Collaborators were also able to choose between imperial and metric units upon station setup. This was also indicated in the column name creating several possible column names for the same measurement. Collaborators were able to set the time upon weather station setup, but the time zone is not indicated in the weather data files requiring collaborators to ensure the time was set properly.

Setup instructions did not include steps to ensure proper siting for the weather station. Therefore, especially in the case of wind speed, observations should be considered microscale and are only representative of the field. Investigators imputed missing dates from the nearest National Weather Service (NWS) Automated Surface Observing Systems (ASOS) airport weather station. To create the cleaned data set investigators converted the data to local time (if necessary) and converted all data to metric units for consistency.

Table 1. Weather data column name and description.

Column Name	Description
Field Location	G2F field-location name
Station ID	Weather station ID number
NWS Network	NWS network identifier created by two letter state abbreviation followed by station type
NWS Station	Nearby National Weather Service (NWS) station used for calibration identified by three or four letter code
Date_key	Year, month, day and time measurement was taken in local time
Month	Month of year measurement was taken in local time
Day	Day of month measurement was taken in local time
Year	Year measurement was taken in local time
Time	Time measurement was taken in local time (Ex: DEH1 is Eastern, IAH1 is Central, etc.)
Temperature [C]	Temperature in degrees Celsius
Dew Point [C]	Dew point temperature in degrees Celsius
Relative Humidity [%]	Relative humidity as a percentage
Solar Radiation [W/m ²]	Incoming solar radiation in Watts per square meter
Rainfall [mm]	Rainfall in millimeters
Wind Speed [m/s]	Wind speed in meters per second
Wind Direction [degrees]	Wind direction in degrees
Wind Gust [m/s]	Largest wind speed over 30-minute period in meters per second
Soil Temperature [C]	Soil temperature in degrees Celsius
Soil Moisture [%VWC]	Soil moisture as a percentage
Soil EC [mS/cm]	Soil electrical conductivity in milliSiemens per centimeter
UV Light [uM/m ² s]	Incoming ultra-violet (UV) radiation in micro-moles per meter-squared seconds
PAR [uM/m ² s]	Photosynthetically active radiation (PAR) in micro-moles per meter-squared seconds